COHORT 16 DSML- MINI PROJECT DOCUMENTATION

Renewable Energy Transition Speed

Date:27/02/2026

DAY 3: ALGORITHM AND FURTHER DATA CLEANING:

Objective:

Prepare raw datasets for modeling and finalize suitable machine learning algorithms for predicting Renewable Energy Transition Speed (SDG-7).

Work Completed:

- Loaded all received datasets into the working environment (VS Code/Jupyter Notebook).
- Checked the number of rows and columns in each dataset.
- Reviewed data types of all features.
- Verified presence of key fields such as Year and Entity/Country.
- Identified missing values across multiple columns.
- Removed rows with excessive null values.
- Applied mean/median imputation where required.
- Forward-filled time-series gaps where logically valid.
- Standardized column names for consistency.
- Removed duplicate records.
- Converted Year to integer data type.
- Converted numerical indicators to float type.
- Filtered irrelevant entities as per project scope.
- Checked and treated outliers using statistical methods.
- Sorted dataset chronologically.
- Merged renewable energy dataset with solar PV price dataset using Year as key.
- Created target variable as Year-over-Year percentage change in renewable share.

Outcome:

A clean, consistent, and merged analytical dataset was generated for model development and saved as clean_energy_data.csv. The dataset is now ready for machine learning implementation.

Group Discussion : Day 3

Figure: Team Meeting – 27th February 2026